

# Robust Controller Synthesis

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**Abstract—**

**Index Terms—**PDEs, Robust Control, LMIs;

## I. INTRODUCTION

This paper uses PN PI operators inspired by the works [2], [5], [12]. Then it IQC extends the results from [1], [4], [11]. [3] also covers robust control using dual iq

## II. NOTATION & PRELIMINARIES

### A. Notation

$L_2^n[a, b]$  denotes the set of  $\mathbb{R}^n$ -valued, Lebesgue square-integrable equivalence classes of functions on the spatial domain  $[a, b] \subset \mathbb{R}$ . The space  $\mathbb{R}L_2^n[a, b]$  denotes the Cartesian product  $\mathbb{R}^{n_1} \times L_2^{n_2}[a, b]$ , where  $\mathbf{n} = \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$ , with inner product

$$\left\langle \begin{bmatrix} x_1 \\ \mathbf{x}_2 \end{bmatrix}, \begin{bmatrix} y_1 \\ \mathbf{y}_2 \end{bmatrix} \right\rangle_{\mathbb{R}L_2} := x_1^\top y_1 + \langle \mathbf{x}_2, \mathbf{y}_2 \rangle_{L_2}.$$

When clear from context, we occasionally omit the subscript and simply write  $\langle \cdot, \cdot \rangle := \langle \cdot, \cdot \rangle_{\mathbb{R}L_2}$ . Since the only spatial domain considered here is  $[a, b]$ , we typically omit the domain and write  $L_2^n$  or  $\mathbb{R}L_2^n$ , further omitting dimensions when clear from context. For a normed space  $X$  and Banach space  $Y$ ,  $\mathcal{L}(X, Y)$  denotes the Banach space of bounded linear operators from  $X$  to  $Y$  with induced norm

$$\|\mathcal{P}\|_{\mathcal{L}(X, Y)} = \sup_{\|\mathbf{x}\|=1} \|\mathcal{P}\mathbf{x}\|_Y,$$

and  $\mathcal{L}(X) := \mathcal{L}(X, X)$ . Our notational convention is to write functions in **bold** (e.g.,  $\mathbf{x}$ , indicating  $\mathbf{x} \in \mathbb{R}L_2$ ) and operators in calligraphic font (e.g.,  $\mathcal{P} \in \mathcal{L}(\mathbb{R}L_2)$ ). For a Hilbert space  $X$  and operator  $\mathcal{A} \in \mathcal{L}(X)$ , the adjoint  $\mathcal{A}^*$  is defined by

$$\langle \mathbf{x}, \mathcal{A}\mathbf{y} \rangle_X = \langle \mathcal{A}^*\mathbf{x}, \mathbf{y} \rangle_X, \quad \forall \mathbf{x}, \mathbf{y} \in X.$$

The space  $L_2^n[0, \infty)$  denotes  $\mathbb{R}^n$ -valued square-integrable signals defined on the temporal domain  $[0, \infty)$ , so that  $x(t) \in \mathbb{R}^n$ . Similarly,  $\mathbb{R}L_2^n[0, \infty)$  denotes  $\mathbb{R}L_2$ -valued square-integrable temporal signals. For suitably differentiable  $\mathbf{x} \in \mathbb{R}L_2^n[0, \infty)$ , we write  $\dot{\mathbf{x}} := \frac{\partial \mathbf{x}}{\partial t}$ . For a Hilbert space  $\mathbf{H}$ , we denote by  $\mathbf{L}_{\mathbf{H}}$  the space of square-integrable functions  $u : [0, \infty) \rightarrow \mathbf{H}$  with inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{L}_{\mathbf{H}}} = \int_0^\infty \langle \mathbf{u}(t), \mathbf{v}(t) \rangle_{\mathbf{H}} dt.$$

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We define the extended space  $\mathbf{L}_{e, \mathbf{H}}$  to consist of functions  $u : [0, \infty) \rightarrow \mathbf{H}$  such that

$$\int_0^T \|\mathbf{u}(t)\|_{\mathbf{H}}^2 dt < \infty \quad \forall T \geq 0.$$

We use  $\mathbf{L}_{e, [a, b]}$  and  $\mathbf{L}_{[a, b]}$  to denote  $\mathbf{L}_{e, \mathbf{H}}$  and  $\mathbf{L}_{\mathbf{H}}$  with  $\mathbf{H} = \mathbb{R}L_2[a, b]$ . For any Hilbert space  $\mathbf{H}$ , the truncation operator  $p_T : \mathbf{L}_{e, \mathbf{H}} \rightarrow \mathbf{L}_{e, \mathbf{H}}$  is defined for  $\mathbf{y} \in \mathbf{L}_{e, \mathbf{H}}$  by

$$(p_T \mathbf{y})(t) = \begin{cases} \mathbf{y}(t), & 0 \leq t \leq T, \\ 0, & \text{otherwise.} \end{cases}$$

such that  $(p_T \mathbf{y})(t) \in \mathbf{L}_{\mathbf{H}}$  for all  $T \geq 0$ .

**Definition 1.** Let  $\mathbf{H}$  and  $\mathbf{G}$  be Hilbert spaces. For all  $T \geq 0$ , an operator  $G : \mathbf{L}_{\mathbf{H}} \rightarrow \mathbf{L}_{\mathbf{G}}$  is:

- **Causal** if  $p_T G = p_T G p_T$ .
- **Bounded on  $\mathbf{L}_{\mathbf{H}}$**  if  $\|G\mathbf{v}\|_{\mathbf{L}_{\mathbf{G}}} \leq C\|\mathbf{v}\|_{\mathbf{L}_{\mathbf{H}}}$  for some  $C \geq 0$  and all  $\mathbf{v} \in \mathbf{L}_{\mathbf{H}}$ .
- **Bounded on  $\mathbf{L}_{e, \mathbf{H}}$**  if  $\|p_T G\mathbf{v}\|_{\mathbf{L}_{\mathbf{G}}} \leq C\|p_T \mathbf{v}\|_{\mathbf{L}_{\mathbf{H}}}$  for some  $C \geq 0$  and all  $\mathbf{v} \in \mathbf{L}_{e, \mathbf{H}}$ .

Finally, we will note some operations. For a given  $\mathcal{K} \in \mathcal{L}(\mathbb{R}L_2^n, \mathbb{R}L_2^m)$ , we define the operator  $\mathcal{M}_{\mathcal{K}} : \mathbf{L}_{e, [a, b]}^{\mathbf{n}} \rightarrow \mathbf{L}_{e, [a, b]}^{\mathbf{m}}$  to compactly represent a multiplication operator. For  $\mathbf{u} = [u_1 \ \cdots \ u_N]^\top$  with  $\mathbf{u} \in \mathbb{R}L_2$ , its action at time  $t$  is given by the block structure of  $\mathcal{K}_{ij}$ :

$$(\mathcal{M}_{\mathcal{K}} \mathbf{u})(t) := \begin{bmatrix} \mathcal{K}_{11} & \cdots & \mathcal{K}_{1n} \\ \vdots & \ddots & \vdots \\ \mathcal{K}_{m1} & \cdots & \mathcal{K}_{mn} \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_n \end{bmatrix} (t).$$

### B. Partial Integral Equations (PIEs)

Partial Integral operators, also known as PI operators, are a class of bounded linear integral operators that are defined jointly on a vector space and Hilbert space.

**Definition 2.** We say  $\mathcal{P} \in \Pi_4^{\mathbf{m} \times \mathbf{n}} \subset \mathcal{L}(\mathbb{R}L_2^{\mathbf{n}}, \mathbb{R}L_2^{\mathbf{m}})$ , denoted by  $\Pi \left[ \begin{array}{c|c} P & Q_1 \\ \hline Q_2 & \{R_0, R_1, R_2\} \end{array} \right]$ , if there exist a matrix  $P$  and matrix polynomials  $Q_1, Q_2, R_0, R_1$ , and  $R_2$  (of compatible dimension) such that,

$$\left[ \left( \Pi \left[ \begin{array}{c|c} P & Q_1 \\ \hline Q_2 & \{R_0, R_1, R_2\} \end{array} \right] \right) \begin{bmatrix} x \\ \mathbf{x} \end{bmatrix} \right] (s) := \begin{bmatrix} Px + \int_a^b Q_1(s) \mathbf{x}(s) ds \\ Q_2(s) x + \mathcal{R} \mathbf{x}(s) \end{bmatrix},$$

$$(\mathcal{R} \mathbf{x})(s) = R_0(s) \mathbf{x}(s) + \int_a^s R_1(s, \theta) \mathbf{x}(\theta) d\theta + \int_s^b R_2(s, \theta) \mathbf{x}(\theta) d\theta.$$

The vector space of PI operators, given compatible dimensions, are closed under composition, addition, adjoint and

concatenation – implying that  $\Pi_4$  is a composition algebra [10].

**Definition 3.** (LPIs) For given PI operators  $\{\mathcal{E}_{i,j}, \mathcal{F}_{i,j}, \mathcal{G}_i\}$  and a convex linear functional  $\mathcal{L}(\cdot)$ , a linear PI inequality is a convex optimization of the following form

$$\begin{aligned} & \min_{P_i, Q_{1i}, Q_{2i}, R_{0i}, R_{1i}, R_{2i}} \mathcal{L}(\{P_i, Q_{1i}, Q_{2i}, R_{0i}, R_{1i}, R_{2i}\}) \\ \text{s.t. } & \sum_{j=1}^K \mathcal{E}_{ij}^* \Pi \begin{bmatrix} P_i & Q_{1i} \\ Q_{2i} & R_i \end{bmatrix} \mathcal{F}_{ij} + \mathcal{G}_i \succeq 0 \end{aligned}$$

Only LMIs are required to solve an optimization problem involving LPIs. These definitions are based on [7]

**Definition 4** (PIE System). Given the PIE system  $G\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$  with PI operators  $\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}$ , and  $\mathcal{D}$  of compatible dimensions, and signals  $w(t) \in \mathbb{R}L_2^{\mathbf{n}_w}[a, b]$ , we say  $\mathbf{x}(t) \in \mathbb{R}L_2[a, b]^{\mathbf{n}_x}$  and  $z(t) \in \mathbb{R}L_2^{\mathbf{n}_z}[a, b]$  satisfy the PIE with initial condition  $\mathbf{x}_0$  if

$$\begin{bmatrix} \partial_t(\mathcal{T}\mathbf{x})(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix}$$

for all  $t \geq 0$  and  $\mathcal{T}\mathbf{x}(0) = \mathcal{T}\mathbf{x}_0$ . Additionally, for signals  $\underline{w}(t) \in \mathbb{R}L_2[a, b]^{\mathbf{n}_z}$  of compatible dimensions, we say  $\underline{\mathbf{x}}(t) \in \mathbb{R}L_2[a, b]^{\mathbf{n}_x}$  and  $\underline{z}(t) \in \mathbb{R}L_2[a, b]^{\mathbf{n}_w}$  satisfy the dual PIE,  $G^\top$ , with initial condition  $\underline{\mathbf{x}}_0$  if

$$\begin{bmatrix} \mathcal{T}^*\underline{\dot{\mathbf{x}}}(t) \\ \underline{z}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & \mathcal{C}^* \\ \mathcal{B}^* & \mathcal{D}^* \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix},$$

for all  $t \geq 0$  and  $\mathcal{T}^*\underline{\mathbf{x}}(0) = \mathcal{T}^*\underline{\mathbf{x}}_0$ .

**Definition 5** (Bounded Internally Stable PIE). Let  $G$  be a PIE system as in Definition 4. We say  $G$  is internally stable and bounded with decay rate  $\alpha > 0$ ,  $M > 0$ , and  $L_2$ -gain  $\gamma > 0$  if

- The homogeneous part  $\partial_t(\mathcal{T}\mathbf{x})(t) = \mathcal{A}\mathbf{x}(t)$ , with  $\mathcal{T}\mathbf{x}(0) = \mathcal{T}\mathbf{x}_0$  is exponentially stable such that, for every admissible initial condition  $\mathbf{x}_0 \in \mathbb{R}L_2$

$$\|\mathcal{T}\mathbf{x}(t)\|_{\mathbf{L}_{[a,b]}} \leq M\|\mathcal{T}\mathbf{x}_0\|_{\mathbf{L}_{[a,b]}} e^{-\alpha t}, \quad \forall t \geq 0.$$

- With zero initial condition  $\mathbf{x}(0) = 0$ , then for any input  $w$  and output  $z$ ,

$$\|p_T z\|_{\mathbf{L}_{[a,b]}} \leq \gamma \|p_T w\|_{\mathbf{L}_{[a,b]}}, \quad \forall T \geq 0, \quad \forall w \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w}.$$

Then  $G \in \mathcal{G}_2^{\mathbf{n}_z \times \mathbf{n}_w}$ . Clearly,  $\mathcal{R}H_\infty \subset \mathcal{G}_2$  [14]. Also  $G^\top \in \mathcal{G}_2$  with the same decay rate,  $\mathbf{L}_{[a,b]}$ -gain, and  $M$  [7].

**Definition 6** (Linear Fractional Transformation (LFT)). Let  $G$  be a causal system or operator with a specified  $2 \times 2$  block partition, and let  $H$  be compatible with either two ports or one port. Write

$$\begin{bmatrix} y_{11} \\ y_{12} \end{bmatrix} = G \begin{bmatrix} u_{11} \\ u_{12} \end{bmatrix}.$$

The Redheffer star products [6] are the following well-posed interconnections.

- If  $H$  has two ports,

$$\begin{bmatrix} y_{21} \\ y_{22} \end{bmatrix} = H \begin{bmatrix} u_{21} \\ u_{22} \end{bmatrix},$$

then

$$\begin{aligned} G \star H : \begin{bmatrix} u_{12} \\ u_{21} \end{bmatrix} &\mapsto \begin{bmatrix} y_{12} \\ y_{21} \end{bmatrix}, & u_{11} = y_{22}, & u_{22} = y_{11}, \\ H \star G : \begin{bmatrix} u_{11} \\ u_{22} \end{bmatrix} &\mapsto \begin{bmatrix} y_{11} \\ y_{22} \end{bmatrix}, & u_{21} = y_{12}, & u_{12} = y_{21}. \end{aligned}$$

- If  $H$  has one port,  $y_H = Hu_H$ , then

$$\begin{aligned} G \star H : u_{12} &\mapsto y_{12}, & u_{11} = y_H, & u_H = y_{11}, \\ H \star G : u_{11} &\mapsto y_{11}, & u_H = y_{12}, & u_{12} = y_H. \end{aligned}$$

### C. PIETOOLS

PIETOOLS is a MATLAB toolbox used for declaration, manipulation and solving LPIs [9]. For learning how to use PIETOOLS and its functionalities visit PIETOOLS homepage.

## III. IQC THEORY

This section introduces the IQC objects used to study stability and performance of interconnected systems in a Hilbert space. We establish the generalized system interconnection, formalize stability and performance, and define the uncertainty and performance IQCs. These definitions provide the foundation for the primal-system analysis in Section IV.

### A. Description of uncertain systems in a Hilbert-space

We consider operators that map external inputs to external outputs through a feedback interconnection between a linear (infinite dimensional) operator  $P$  and  $\Delta$ . This interconnection structure is formally defined as follows

**Definition 7** (Generalized Plant). Given operators  $P_\Delta : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{w_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{z_\Delta}}$ ,  $P_{\Delta p} : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{w_p}} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{z_\Delta}}$ ,  $P_{p\Delta} : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{w_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{z_p}}$ ,  $P_p : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{w_p}} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{z_p}}$ , and  $\Delta : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{z_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{z_\Delta}}$ , define

$$\begin{bmatrix} z_\Delta \\ z_p \end{bmatrix} = P \begin{bmatrix} w_\Delta \\ w_p \end{bmatrix} = \begin{bmatrix} P_\Delta & P_{\Delta p} \\ P_{p\Delta} & P_p \end{bmatrix} \begin{bmatrix} w_\Delta \\ w_p \end{bmatrix}, \quad w_\Delta = \Delta z_\Delta. \quad (1)$$

for input  $w_p \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{w_p}}$  and output  $z_p \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{z_p}}$ , there exist internal signals  $z_\Delta \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{z_\Delta}}$  and  $w_\Delta \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_{w_\Delta}}$  satisfying the interconnected system  $z_p = (P \star \Delta)w_p$ .

The interconnection structure described above captures the feedback interaction between the nominal system  $P$  and the uncertainty  $\Delta$ , where  $w_p$  and  $z_p$  represent external inputs and performance outputs, respectively, while  $w_\Delta$  and  $z_\Delta$  form the internal feedback loop through the uncertainty. Before analyzing stability and performance, we must first ensure the well-posedness of the interconnection, such that solutions exist, are unique, and depend causally on inputs.

**Definition 8.** We say that the uncertain interconnection  $P \star \Delta$  is well-posed if:

- **Existence and Uniqueness:** For any  $w \in \mathbf{L}_{e,[a,b]}$ , there exist unique  $z \in \mathbf{L}_{e,[a,b]}$ ,  $z_\Delta \in \mathbf{L}_{e,[a,b]}$  and  $w_\Delta \in \mathbf{L}_{e,[a,b]}$  such that,  $z$ ,  $w$ ,  $z_\Delta$ ,  $w_\Delta$  satisfy the interconnection.

- **Causality:** If  $z_\Delta, w_\Delta$  and  $\hat{z}_\Delta, \hat{w}_\Delta$  satisfy the interconnection, then  $p_T(z_\Delta - \hat{z}_\Delta) = 0$  and  $p_T(w_\Delta - \hat{w}_\Delta) = 0$ , for all  $T \geq 0$ .

Having established the class of uncertain systems and their well-posedness, we now develop a framework to guarantee robust stability and performance for the class of systems defined in Definition 7.

### B. Robust stability and performance

Traditional control system design often treats stability and performance shaping as separate objectives with distinct constraints. However, Integral Quadratic Constraints (IQCs) provide a unified framework where these trade-offs can be captured within a single constraint [11]. We now define what it means for the system to remain stable despite uncertainties. Robust stability requires that the system is well-posed for all admissible uncertainties and that the input-output map remains bounded. This involves removal of the performance channels in (1), and introducing auxiliary signals  $z_0$  and  $w_0$ .

**Definition 9.** *The interconnection*

$$z_\Delta = P_\Delta w_\Delta + z_0, \quad w_\Delta = \Delta z_\Delta + w_0,$$

with  $z_0, w_0 \in \mathbf{L}_{e,[a,b]}$ , is said to be **robustly stable** if:

- 1) The uncertain system is well-posed for all  $\Delta \in \Delta$ .
- 2) The map from  $\begin{bmatrix} z_0 \\ w_0 \end{bmatrix}$  to  $\begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}$  is bounded on  $\mathbf{L}_{e,[a,b]}$  for all  $\Delta \in \Delta$ .

Within the IQC framework, the stability of an interconnection  $P \star \Delta$  is certified by verifying that the uncertainty operator  $\Delta$  satisfies a quadratic constraint defined by an appropriate multiplier. This leads to the following formal definition

**Definition 10.** An uncertainty operator  $\Delta : \mathbf{L}_{e,[a,b]}^{n_{z_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}}$  is said to satisfy the hard uncertainty IQC  $(\Psi_\Delta, \mathcal{M}_{\mathcal{V}_\Delta})$  defined by the dynamic operator  $\Psi_\Delta : \mathbf{L}_{e,[a,b]}^{n_{z_\Delta} + n_{w_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z_\Delta} + n_{w_\Delta}}$  multiplication operator  $\mathcal{M}_{\mathcal{V}_\Delta}$ , parameterized by  $\mathcal{V}_\Delta = \mathcal{V}_\Delta^* \in \mathcal{L}(\mathbb{R}L_2, \mathbb{R}L_2)$  if the inequality holds for all  $z_\Delta \in \mathbf{L}_{e,[a,b]}$  and all  $T \geq 0$ ,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \geq 0.$$

Moreover, let  $\mathcal{V}_\Delta\{\mathcal{Q}_\Delta, \mathcal{S}_\Delta, \mathcal{R}_\Delta\}$  be parameterized as,

$$\mathcal{V}_\Delta = \begin{bmatrix} \mathcal{Q}_\Delta & \mathcal{S}_\Delta \\ \mathcal{S}_\Delta^* & \mathcal{R}_\Delta \end{bmatrix}$$

Beyond stability, we are interested in ensuring that the system meets certain performance specifications in the presence of uncertainty. Robust performance extends the notion of robust stability by additionally requiring that a performance metric is satisfied.

**Definition 11.** *Robust performance* is achieved if the interconnection  $P \star \Delta$  is **robustly stable** for some hard performance IQC  $(\Psi_p, \mathcal{M}_{\mathcal{V}_p})$  defined by the dynamic operator

$\Psi_p : \mathbf{L}_{e,[a,b]}^{n_{z_p} + n_{w_p}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z_p} + n_{w_p}}$  and multiplication operator  $\mathcal{M}_{\mathcal{V}_p}$ , parameterized by  $\mathcal{V}_p = \mathcal{V}_p^* \in \mathcal{L}(\mathbb{R}L_2, \mathbb{R}L_2)$  with

$$\left\langle p_T \Psi_p \begin{bmatrix} P_p \Delta \\ 0 \end{bmatrix} w_p, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} P_p \Delta \\ 0 \end{bmatrix} w_p \right\rangle_{\mathbf{L}_{[a,b]}} \geq 0.$$

for all  $w_p \in \mathbf{L}_{e,[a,b]}^{n_{w_p}}$ . Then there exists a sufficiently small  $\epsilon > 0$  such that, for all such  $w_p$  and all  $T \geq 0$ ,

$$\left\langle p_T \Psi_p \begin{bmatrix} P \star \Delta \\ I \end{bmatrix} w_p, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} P \star \Delta \\ I \end{bmatrix} w_p \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_p\|_{\mathbf{L}_{[a,b]}}^2.$$

Moreover, let  $\mathcal{V}_p\{\mathcal{Q}_p, \mathcal{S}_p, \mathcal{R}_p\}$  be parameterized as,

$$\mathcal{V}_p = \begin{bmatrix} \mathcal{Q}_p & \mathcal{S}_p \\ \mathcal{S}_p^* & \mathcal{R}_p \end{bmatrix}$$

In this framework, the multipliers  $\mathcal{M}_{\mathcal{V}_\Delta}$  and  $\mathcal{M}_{\mathcal{V}_p}$  serve distinct but complementary roles. The hard uncertainty IQC  $(\Psi_\Delta, \mathcal{M}_{\mathcal{V}_\Delta})$  characterizes the behavior of uncertainties, nonlinearities, and other perturbations captured by the operator  $\Delta$ , ensuring that the IQC is satisfied for the class of admissible disturbances. In contrast, the hard performance IQC  $(\Psi_p, \mathcal{M}_{\mathcal{V}_p})$  encodes performance requirements for the nominal system, such as  $L_2$ -gain, passivity conditions, or other desired input-output properties. Together, these multipliers enable simultaneous verification of robust stability against perturbations and achievement of specified performance objectives. This section applies the IQC framework to the primal system and gives a certificate for simultaneous robust stability and performance. For brevity, we write the hard IQC as

$$\left\langle p_T \Psi \begin{bmatrix} Pw \\ w \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}} \Psi \begin{bmatrix} Pw \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2.$$

Let  $\Psi_\Delta$  and  $\Psi_p$  have a  $2 \times 2$  block partition and define  $\Psi_{ij} := \text{diag}((\Psi_\Delta)_{ij}, (\Psi_p)_{ij})$  for  $i, j \in \{1, 2\}$ . Let  $w^\top = [w^\top \ w_p^\top]$  and  $z^\top = [z_\Delta^\top \ z_p^\top]$ . Moreover, define  $\mathcal{Q} = \text{diag}(\mathcal{Q}_\Delta, \mathcal{Q}_p)$ ,  $\mathcal{S} = \text{diag}(\mathcal{S}_\Delta, \mathcal{S}_p)$ , and  $\mathcal{R} = \text{diag}(\mathcal{R}_\Delta, \mathcal{R}_p)$ , then  $\mathcal{V}\{\mathcal{Q}, \mathcal{S}, \mathcal{R}\}$ .

**Theorem 1** (Robust Stability and Performance). *Let  $P, \Delta \in \Delta$  be given, and there exists  $\Psi_\Delta, \mathcal{M}_{\mathcal{V}_\Delta}$ , such that,*

- The interconnection  $P \star \Delta$  is well-posed for all  $\Delta \in \Delta$
- The IQC in Definition 10 is satisfied for all  $\Delta \in \Delta$ .

*Then the interconnection  $P \star \Delta$  is **robustly stable** and **robust performance** is achieved with respect to some  $\Psi_p$  and  $\mathcal{M}_{\mathcal{V}_p}$ , in Definition 11, if,*

$$\left\langle p_T \Psi \begin{bmatrix} Pw \\ w \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}} \Psi \begin{bmatrix} Pw \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2 \quad (2)$$

for all  $w_\Delta \in \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}}$ ,  $w_p \in \mathbf{L}_{e,[a,b]}^{n_{w_p}}$  and  $T \geq 0$ .

*Proof.* See Appendix A.1. ■

Theorem 1 establishes a framework for deriving Linear PI Inequalities (LPIs), which are used in assessing the robust performance of a system.

$$\begin{aligned}
P &:= \begin{cases} \begin{bmatrix} \mathcal{T}_P \dot{\mathbf{x}}_P(t) \\ z(t) \\ y_P(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P & \mathcal{B}_P & \mathcal{B}_u \\ \mathcal{C}_P & \mathcal{D}_P & \mathcal{D}_{zu} \\ I & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_P(t) \\ w(t) \\ u(t) \end{bmatrix} \\ K &:= \begin{cases} u(t) = [\mathcal{K}_P \quad \mathcal{K}_\Theta] \begin{bmatrix} \mathbf{x}_P(t) \\ \mathbf{x}_\Theta(t) \end{bmatrix} \end{cases} \\ \Theta &:= \begin{cases} \begin{bmatrix} \mathcal{T}_\Theta \dot{\mathbf{x}}_\Theta(t) \\ \tilde{v}(t) \\ y_\Theta(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \\ I & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Theta(t) \\ v(t) \end{bmatrix} \end{cases} \\ \Theta \begin{bmatrix} K \star P \\ I \end{bmatrix} &:= \begin{cases} \begin{bmatrix} \mathcal{T}_P \dot{\mathbf{x}}_P(t) \\ \mathcal{T}_\Theta \dot{\mathbf{x}}_\Theta(t) \\ \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P + \mathcal{B}_u \mathcal{K}_P & \mathcal{B}_u \mathcal{K}_\Theta & \mathcal{B}_P \\ \mathcal{B}_1(\mathcal{C}_P + \mathcal{D}_{zu} \mathcal{K}_P) & \mathcal{A} + \mathcal{B}_1 \mathcal{D}_{zu} \mathcal{K}_\Theta & \mathcal{B}_1 \mathcal{D}_P + \mathcal{B}_2 \\ \mathcal{D}_{11}(\mathcal{C}_P + \mathcal{D}_{zu} \mathcal{K}_P) & \mathcal{C}_1 + \mathcal{D}_{11} \mathcal{D}_{zu} \mathcal{K}_\Theta & \mathcal{D}_{11} \mathcal{D}_P + \mathcal{D}_{12} \\ \mathcal{D}_{21}(\mathcal{C}_P + \mathcal{D}_{zu} \mathcal{K}_P) & \mathcal{C}_2 + \mathcal{D}_{21} \mathcal{D}_{zu} \mathcal{K}_\Theta & \mathcal{D}_{21} \mathcal{D}_P + \mathcal{D}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{x}_P(t) \\ \mathbf{x}_\Theta(t) \\ w(t) \end{bmatrix} \end{cases} \end{cases} \quad (3)
\end{aligned}$$

Fig. 1. Primal component realizations and full augmented interconnection for state feedback with dynamic filtering.

#### IV. CONTROLLER ANALYSIS (PRIMAL SYSTEM)

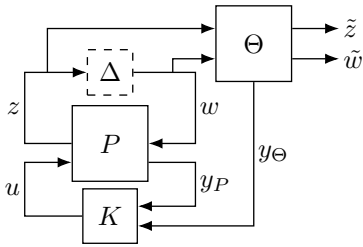


Fig. 2. Primal controller interconnection.

#### V. DUALITY

In this work we assume that the generalized plant  $P$  is a PIE system, that admits a state-space representation. This work builds on the duality results in [7] which shows that the state-feedback controller results in a convex optimization problem. We find that this duality result can be extended to the case of dynamic filters, where we want to know the properties of the dual dynamic filter. This is particularly useful when examining the effect of uncertainties on the system when performing state-feedback controller synthesis, when using IQCs. The works from Seiler and Venkatarman [8], [12] show that, after  $J$ -spectral factorization, the dual of a dynamic filter can be found using the Potapov-Ginzburg transform. This section extends these results to PIE systems, and shows that the dual of a PIE system with a dynamic filter can be found using the Potapov-Ginzburg transform, identical to the work from Venkatarman.

##### A. Duality under Dynamic Filtering

We begin by restating the main theorem from [7].

**Theorem 2** (Dual  $L_2$ -gain [7, Thm. 9]). *For a PIE system with corresponding PI operators  $P\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$ , and some  $\gamma > 0$ , the following statements are equivalent.*

- 1) *For any  $w \in \mathbf{L}_{e,[a,b]}^{n_w}$ , and  $\mathbf{x}(0) = 0$ ,  $P$  satisfies  $\|p_T z\|_{\mathbf{L}} \leq \gamma \|p_T w\|_{\mathbf{L}}$  for all  $T > 0$ .*
- 2) *For any  $\underline{w} \in \mathbf{L}_{e,[a,b]}^{n_z}$ , and  $\underline{\mathbf{x}}(0) = 0$ ,  $P^\top$  satisfies  $\|p_T \underline{z}\|_{\mathbf{L}} \leq \gamma \|p_T \underline{w}\|_{\mathbf{L}}$  for all  $T > 0$ .*

Relating an IQC to this gain duality requires two transformations. First, write  $\begin{bmatrix} \tilde{z} \\ \tilde{w} \end{bmatrix} := \Theta \begin{bmatrix} z \\ w \end{bmatrix}$ . A  $J$ -spectral factorization gives, for every  $T \geq 0$ ,

$$\begin{aligned}
&\left\langle p_T \Psi \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_V \Psi \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \\
&= \|p_T \tilde{z}\|_{\mathbf{L}_{[a,b]}}^2 - \|p_T \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2.
\end{aligned}$$

Thus the IQC quadratic form becomes the difference between the filtered output and input signals.

**Definition 12** ( $J$ -Spectral Factorization and Dual Filter [12]). *Let  $(\Psi, \mathcal{M}_V)$  be a hard IQC, let*

$$J_{n_z, n_w} := \begin{bmatrix} I_{n_z} & 0 \\ 0 & -I_{n_w} \end{bmatrix},$$

*and let  $\Theta, \Theta^{-1}, \Theta_{22}, \Theta_{22}^{-1}$  be causal bounded linear operators. The pair  $(\Theta, J_{n_z, n_w})$  is a  $J$ -spectral factorization of  $(\Psi, \mathcal{M}_V)$  if,*

$$\begin{aligned}
&\left\langle p_T \Psi \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_V \Psi \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \\
&= \left\langle p_T \Theta \begin{bmatrix} z \\ w \end{bmatrix}, p_T J_{n_z, n_w} \Theta \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}}.
\end{aligned}$$

*Define the matrix  $\hat{J} := \begin{bmatrix} 0_{n_z \times n_w} & I_{n_z} \\ -I_{n_w} & 0_{n_w \times n_z} \end{bmatrix}$ . The dual filter associated with  $\Theta$  is defined by*

$$\mathbf{D}(\Theta) := (\hat{J}^* \Theta^\top \hat{J})^{-1}.$$

Give the primal inner product,

$$\mathcal{I} := \left\langle p_T \Theta \begin{bmatrix} Pw \\ w \end{bmatrix}, p_T J_{n_z, n_w} \Theta \begin{bmatrix} Pw \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}}$$

and corresponding dual iq

$$\underline{\mathcal{I}} := \left\langle p_T \mathbf{D}(\Theta) \begin{bmatrix} P^\top \underline{w} \\ \underline{w} \end{bmatrix}, p_T J_{n_w, n_z} \mathbf{D}(\Theta) \begin{bmatrix} P^\top \underline{w} \\ \underline{w} \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}}$$

Although the following theorem can be derived using state-space realizations, the proof is cumbersome; for conciseness and clarity, we instead use an operator-based derivation.

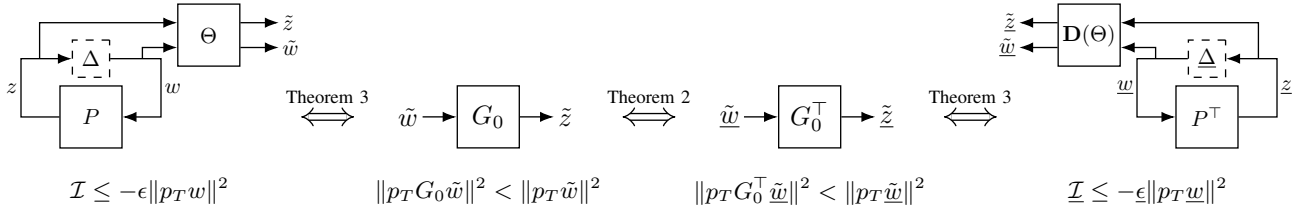


Fig. 3. Equivalence chain for primal and dual dynamic filtering.

**Theorem 3.** Let  $P$  satisfy Definition 7 and let  $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$  be a hard  $J$ -spectral factorization. Finally, assume  $(\Theta_{21}P + \Theta_{22})$  has a bounded causal inverse. Define

$$G_0 = (\Theta_{11}P + \Theta_{12})(\Theta_{21}P + \Theta_{22})^{-1}$$

then the following statements are equivalent,

- 1) There exists a  $0 < \rho < 1$  such that  $\|p_T G_0 \tilde{w}\|_{\mathbf{L}_{[a,b]}} \leq \rho \|p_T \tilde{w}\|_{\mathbf{L}_{[a,b]}}$  for every  $\tilde{w}$  and  $T \geq 0$ .
- 2) There exists some  $\epsilon > 0$  such that  $\mathcal{I} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2$  for every  $w$  and  $T \geq 0$ .
- 3) There exists some  $\underline{\epsilon} > 0$  such that  $\underline{\mathcal{I}} \leq -\underline{\epsilon} \|p_T \underline{w}\|_{\mathbf{L}_{[a,b]}}^2$  for every  $\underline{w}$  and  $T \geq 0$ .

*Proof.* See Appendix A.2. ■

**Remark 1** (On the invertibility assumption). *Bounded causal invertibility of  $\Gamma := \Theta_{21}P + \Theta_{22}$  can be certified using the bounded-real lemma. First verify that the feedthrough of  $\Gamma$  has a bounded causal inverse and use it to construct a PIE realization of  $\Gamma^{-1}$ . Feasibility of the bounded-real KYP inequality for this realization, with a coercive Lyapunov operator and a strict decay margin, certifies exponential internal stability and finite  $L_2$ -gain of the causal inverse.*

**Corollary 1.** Let the realization of  $P$  satisfy Definition 7. Let  $\Delta \in \mathbf{\Delta}$  and there exist *either*, a primal hard IQC  $(\Psi, \mathcal{M}_{\mathcal{V}})$ , with  $\mathcal{V} \in \Pi_4$ , or a dual hard IQC  $(\underline{\Psi}, \mathcal{M}_{\underline{\mathcal{V}}})$ , with  $\underline{\mathcal{V}} \in \Pi_4$ , such that,

- The interconnection  $P \star \Delta$  is well-posed for all  $\Delta \in \mathbf{\Delta}$ .
- The IQC in Definition 10 is satisfied for all  $\Delta \in \mathbf{\Delta}$ .
- The primal hard IQC  $(\Psi, \mathcal{M}_{\mathcal{V}})$  has a hard  $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$  factorization.

Suppose that *either* of the two statements hold for some  $\mathcal{P} = \mathcal{P}^* \succeq 0 \in \Pi_4$ .

- $$\begin{bmatrix} \mathcal{T}^* \mathcal{P} \mathcal{A} + (*) & \mathcal{T}^* \mathcal{P} \mathcal{B} \\ \mathcal{B}^* \mathcal{P} \mathcal{T} & \epsilon I \end{bmatrix} + \begin{bmatrix} \mathcal{C}^* \\ \mathcal{D}^* \end{bmatrix} \mathcal{V} \begin{bmatrix} \mathcal{C} & \mathcal{D} \end{bmatrix} \preceq 0$$
- $$\begin{bmatrix} \mathcal{T}^* \mathcal{P} \mathcal{A} + (*) & \mathcal{T}^* \mathcal{P} \mathcal{B} \\ \mathcal{B}^* \mathcal{P} \mathcal{T} & \epsilon I \end{bmatrix} + \begin{bmatrix} \mathcal{C}^* \\ \mathcal{D}^* \end{bmatrix} \underline{\mathcal{V}} \begin{bmatrix} \mathcal{C} & \mathcal{D} \end{bmatrix} \preceq 0 \quad (4)$$

Then for any  $w, \underline{w} \in \mathbf{L}_{e,[a,b]}$ , satisfies either

$$\Psi \begin{bmatrix} P \\ I \end{bmatrix} w := \left\{ \begin{bmatrix} \mathcal{T} \dot{\mathbf{x}}(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix} \right\}$$

or

$$\underline{\Psi} \begin{bmatrix} P^T \\ I \end{bmatrix} \underline{w} := \left\{ \begin{bmatrix} \mathcal{T} \dot{\underline{\mathbf{x}}}(t) \\ \underline{v}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix} \right\} \quad (5)$$

Then the interconnection  $P \star \Delta$  with  $\mathbf{x}(0) = 0$  is robustly performing.

*Proof.* See Appendix A.3 ■

**Remark 2.** Note that we do not require information about the dual uncertainty  $\underline{\Delta}$ , only the primal uncertainty  $\Delta$ .

## VI. CONVEX CONTROLLER SYNTHESIS (DUAL SYSTEM)

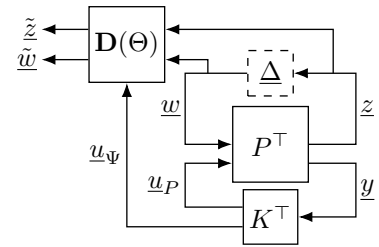


Fig. 5. Dual controller interconnection.

Theorem 1 provides a robust-performance certificate for PIE systems in terms of the dual IQC. For analysis, this merely reformulates the primal certificate, since primal–dual equivalence preserves feasibility. The benefit of the dual formulation therefore appears in controller synthesis. Both controller constructions below use the same synthesis LPI, obtained by applying (4) to the corresponding closed-loop PIE. The choice of dual uncertainty filter changes only the realization of the controller recovered from this LPI. A related application of IQC duality to convex feedforward controller synthesis is given by Venkataraman and Seiler [12].

### A. Primal and Dual Systems

Let  $P$ , with zero initial condition  $\mathbf{x}(0) = 0$ , have input  $w$ , control input  $u$ , and output  $z$ . Its primal and dual PIE state-space realizations are summarized in the corresponding system-component boxes. If  $\Psi \neq I$ , assume that  $\mathbf{D}(\Psi)$  exists and that  $z, w$  are measurable.

### B. State Feedback LPI

**Corollary 2.** Let the realization of  $P$  satisfy Definition 7. Let  $\Delta \in \mathbf{\Delta}$ , there exists a primal hard IQC  $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$ , such that,

- The interconnection  $P \star \Delta$  is well-posed for all  $\Delta \in \mathbf{\Delta}$ .
- The IQC in Definition 10 is satisfied for all  $\Delta \in \mathbf{\Delta}$ .

Moreover there exist,  $\mathcal{Z}, \mathcal{P} \in \Pi_4$  with  $\mathcal{P} = \mathcal{P}^* \succ 0$ , such that the following statement holds

$$\begin{bmatrix} \mathcal{T}^* \mathcal{P} \mathcal{A}_0 + \mathcal{T}^* \mathcal{Z} [\mathcal{B}_u^* & 0] + (*) & \mathcal{T}^* \mathcal{P} \mathcal{B}_0 + \mathcal{T}^* \mathcal{Z} \mathcal{D}_{zu}^* \\ (*) & \underline{\epsilon} I \end{bmatrix} + \begin{bmatrix} \mathcal{C}^* \\ \mathcal{D}^* \end{bmatrix} \underline{\mathcal{V}} \begin{bmatrix} \mathcal{C} & \mathcal{D} \end{bmatrix} \preceq 0. \quad (7)$$

$$\begin{aligned}
P^\top &:= \left\{ \begin{bmatrix} \mathcal{T}_P^* \dot{\mathbf{x}}_P(t) \\ \underline{z}(t) \\ \underline{y}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P^* & \mathcal{C}_P^* & I \\ \mathcal{B}_P^* & \mathcal{D}_P^* & 0 \\ \mathcal{B}_u^* & \mathcal{D}_{zu}^* & 0 \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}_P(t) \\ \underline{w}(t) \\ \underline{u}_P(t) \end{bmatrix} \right. \\
K^\top &:= \left\{ \begin{bmatrix} \underline{u}_P(t) \\ \underline{u}_\Theta(t) \end{bmatrix} = \begin{bmatrix} \mathcal{K}_P^* \\ \mathcal{K}_\Theta^* \end{bmatrix} \underline{y}(t) \right. \\
\mathbf{D}(\Theta) &:= \left\{ \begin{bmatrix} \mathcal{T}^* \dot{\mathbf{x}}_\Theta(t) \\ \underline{\tilde{v}}(t) \end{bmatrix} = \begin{bmatrix} \underline{\mathcal{A}} & \underline{\mathcal{B}} & I \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} & 0 \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}_\Theta(t) \\ \underline{v}(t) \\ \underline{u}_\Theta(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* - \mathcal{C}^* \hat{\mathbf{J}} \mathbf{D}(\mathcal{D}) \hat{\mathbf{J}}^* \mathcal{B}^*, & \mathcal{C}^* \hat{\mathbf{J}} \mathbf{D}(\mathcal{D}) & I \\ -\mathbf{D}(\mathcal{D}) \hat{\mathbf{J}}^* \mathcal{B}^* & \mathbf{D}(\mathcal{D}) & 0 \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}_\Theta(t) \\ \underline{v}(t) \\ \underline{u}_\Theta(t) \end{bmatrix} \right. \\
\mathbf{D}(\Theta) \begin{bmatrix} (K^\star P)^\top \\ I \end{bmatrix} &:= \left\{ \begin{bmatrix} \mathcal{T}_P^* \dot{\mathbf{x}}_P(t) \\ \mathcal{T}^* \dot{\mathbf{x}}_\Theta(t) \\ \underline{\tilde{z}}(t) \\ \underline{\tilde{w}}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P^* + \mathcal{K}_P^* \mathcal{B}_u^* & 0 & \mathcal{C}_P^* + \mathcal{K}_P^* \mathcal{D}_{zu}^* \\ \underline{\mathcal{B}}_1 \mathcal{B}_P^* + \mathcal{K}_\Theta^* \mathcal{B}_u^* & \underline{\mathcal{A}} & \underline{\mathcal{B}}_1 \mathcal{D}_P^* + \mathcal{B}_2 + \mathcal{K}_\Theta^* \mathcal{D}_{zu}^* \\ \underline{\mathcal{D}}_{11} \mathcal{B}_P^* & \underline{\mathcal{C}}_1 & \underline{\mathcal{D}}_{11} \mathcal{D}_P^* + \underline{\mathcal{D}}_{12} \\ \underline{\mathcal{D}}_{21} \mathcal{B}_P^* & \underline{\mathcal{C}}_2 & \underline{\mathcal{D}}_{21} \mathcal{D}_P^* + \underline{\mathcal{D}}_{22} \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}_P(t) \\ \underline{\mathbf{x}}_\Theta(t) \\ \underline{w}(t) \end{bmatrix} \right. \quad (6)
\end{aligned}$$

Fig. 4. Dual component realizations and full augmented interconnection for state feedback with dynamic filtering, expressed in terms of the primal PI operators.

where

$$\mathbf{D}(\Psi) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} := \left\{ \begin{bmatrix} \mathcal{T}^* \dot{\underline{x}}(t) \\ \underline{v}(t) \end{bmatrix} = \begin{bmatrix} \underline{\mathcal{A}}_0 & \underline{\mathcal{B}}_0 \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \begin{bmatrix} \underline{\xi}(t) \\ \underline{w}(t) \end{bmatrix} \right. \quad (8)$$

and

$$\mathcal{Z} := \mathcal{P} \begin{bmatrix} \mathcal{K}_P^* \\ \mathcal{K}_\Psi^* \end{bmatrix},$$

Then the interconnection  $K \star P \star \Delta$  with  $\mathbf{x}(0) = 0$  is robustly stable and robust performance is achieved.

### C. $\underline{\Psi}_\Delta = I$ controller realization

When  $\Psi_\Delta = I$  and  $\underline{\Psi}_\Delta = I$ , no multiplier state is required in the controller. For a state-feedback operator  $\mathcal{K}_P$ , set  $u(t) = \mathcal{K}_P \mathbf{x}_P(t)$ . The resulting closed-loop PIE  $\mathcal{K} \star \mathcal{P}$  is

$$\begin{bmatrix} \partial_t(\mathcal{T}_P \mathbf{x}_P(t)) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_P + \mathcal{B}_u \mathcal{K}_P & \mathcal{B}_P \\ \mathcal{C}_P + \mathcal{D}_{zu} \mathcal{K}_P & \mathcal{D}_P \end{bmatrix} \begin{bmatrix} \mathbf{x}_P(t) \\ w(t) \end{bmatrix}.$$

### D. Dynamic-filter controller realization

Following [13], assume that the uncertainty  $\Delta$  is known and that  $z_\Delta$ ,  $w_\Delta = \Delta z_\Delta$ , and  $w_p$  are available to the controller. For  $\underline{\Psi}_\Delta \neq I$ , find the realization of  $\Psi_\Delta = \mathbf{D}(\underline{\Psi}_\Delta)$ ,

$$\begin{bmatrix} \mathcal{T}_{\Psi_\Delta} \dot{\mathbf{x}}_{\Psi_\Delta}(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_{\Psi_\Delta} & \mathcal{B}_{\Psi_\Delta}^z & \mathcal{B}_{\Psi_\Delta}^w \\ I & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_{\Psi_\Delta}(t) \\ z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}.$$

We impose the zero initial condition  $\mathbf{x}_{\Psi_\Delta}(0) = 0$ . The second block row gives  $y(t) = \mathbf{x}_{\Psi_\Delta}(t)$ . The complete controller has the PIE realization

$$\begin{bmatrix} \mathcal{T}_{\Psi_\Delta} \dot{\mathbf{x}}_{\Psi_\Delta}(t) \\ u(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_{\Psi_\Delta} & 0 & \mathcal{B}_{\Psi_\Delta}^z & \mathcal{B}_{\Psi_\Delta}^w \\ \mathcal{K}_\Psi & \mathcal{K}_P & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_{\Psi_\Delta}(t) \\ \mathbf{x}(t) \\ z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}.$$

Hence  $\mathbf{x}_{\Psi_\Delta}$  is the controller state,  $\mathbf{x}$ , and relevant,  $z_\Delta$ , and  $w_\Delta$  are its external inputs, and  $u$  is its output.

## VII. NUMERICAL EXAMPLES

We will consider examples from [7] but add uncertainty. Moreover, an additional example is considered namely the problem of oscillations in the Subthalamic Nucleus.

**Example 1** (Reaction-Diffusion Equation). *Consider boundary control of an unstable uncertain reaction-diffusion equation:*

$$\begin{aligned}
\dot{\mathbf{v}}(t, s) &= \partial_s^2 \mathbf{v}(t, s) + 5\mathbf{v}(t, s) + \frac{8}{5}w_\Delta(t, s) + s(1-s)w_p(t), \\
\mathbf{v}(t, 0) &= 0, \quad \partial_s \mathbf{v}(t, 1) = x(t), \quad \dot{x}(t) = u(t), \\
z_\Delta(t, s) &= \mathbf{v}(t, s), \quad z_p(t) = \int_0^1 \mathbf{v}(t, s) ds,
\end{aligned}$$

The PIE realization of  $P$  has state  $\mathbf{x}(t) = \partial_s^2 \mathbf{v}(t)$ ,  $(\Delta z_\Delta) = \delta z_\Delta$  with  $|\delta| < 1$  and PI operators,

$$\begin{aligned}
\mathcal{T}_P &= \Pi \left[ \begin{array}{c|c} 1 & 0 \\ \hline s & \{0, -\theta, -s\} \end{array} \right], \quad \mathcal{A}_P = \Pi \left[ \begin{array}{c|c} 0 & 0 \\ \hline 5s & \{1, -5\theta, -5s\} \end{array} \right] \\
\mathcal{B}_P &= \Pi \left[ \begin{array}{c|c} 0 & [0 \ 0] \\ \hline 0.1 & \left\{ \left[ \frac{4}{5} \ \frac{4}{5} \right], [0 \ 0], [0 \ 0] \right\} \end{array} \right] \\
\mathcal{B}_u &= \Pi \left[ \begin{array}{c|c} 1 & \emptyset \\ \hline 0 & \{\emptyset\} \end{array} \right] \\
\mathcal{C}_P &= \Pi \left[ \begin{array}{c|c} \left[ \begin{array}{c} 1 \\ \frac{1}{2} \\ 0 \end{array} \right] & \left[ \begin{array}{c} 0 \\ \frac{1}{2}s^2 - s \\ 0 \end{array} \right] \\ \hline \left[ \begin{array}{c} 0 \\ s \end{array} \right] & \left\{ \left[ \begin{array}{c} 0 \\ 1 \end{array} \right], \left[ \begin{array}{c} 0 \\ -\theta \end{array} \right], \left[ \begin{array}{c} 0 \\ -s \end{array} \right] \right\} \end{array} \right]
\end{aligned}$$

For this example we used  $\underline{\Delta} = \Delta^* = \Delta$  instead of relying on factorization.  $\underline{\Delta} = \Delta^* = \Delta$  implies that the same multiplier structure can be used on the primal and dual side. [11] gives the following primal multiplier structure  $\mathcal{V}_\Delta\{\mathcal{Q}_\Delta, \mathcal{S}_\Delta, -\mathcal{Q}_\Delta\}$ , with  $\mathcal{Q}_\Delta^* = \mathcal{Q}_\Delta \succeq 0 \in \Pi_4$  and  $\mathcal{S}_\Delta^* = -\mathcal{S}_\Delta \in \Pi_4$  with  $\Psi_\Delta = I$  as  $\Delta$  is assumed to be unmeasurable. The dual operators associated with the uncertainty have identical properties. Then define the primal  $\mathcal{V}_p\{I_2, 0, -\gamma^2\}$  and dual  $\underline{\mathcal{V}}_p\{1, 0, -\gamma^2 I_2\}$ , i.e. we are looking for the induced  $L_2$ -gain

on the primal and dual side. Synthesizing a state feedback controller corresponding to Theorem 2 yields,

$$u(t) = -247.26x(t) + \int_0^1 K(s) \partial_s^2 \mathbf{v}(t, s) ds,$$

$$K(s) = 1.93s^{10} - 20.76s^9 + 81.61s^8 - 165.38s^7 + 191.14s^6 - 94.18s^5 + 0.20s^4 - 102.86s^3 - 14.63s^2 + 335.30s - 0.004$$

Synthesis yielded a gain of 0.14242, analysis yielded  $\gamma = 0.13972$  and  $\underline{\gamma} = 0.13979$ .

#### A. Sector and Slope restricted uncertainty

**Lemma 1** (Dual Sector Bound IQC). *Suppose that for  $z_\Delta \in \mathbf{L}_e, [a, b]$ ,  $(\Delta z_\Delta)(s, t) = \phi(z_\Delta(s, t))$  for all  $s \in [a, b]$  and  $t > 0$ , where  $\phi$  satisfies*

$$\alpha z^2 \leq z\phi(z) < \beta z^2 \quad \text{for all } z \in \mathbb{R}$$

with  $\alpha, \beta \in \mathbb{R}$ . Then  $\Delta$  satisfies the dual Hard IQC defined by  $\underline{\Psi}_\Delta = I$  and

$$\underline{\mathcal{V}}_\Delta = \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}^* \begin{bmatrix} 0 & \underline{\mathcal{S}} \\ \underline{\mathcal{S}} & 0 \end{bmatrix} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix} \quad (9)$$

for and  $\underline{\mathcal{S}} \in \Pi_3$  with

$$(\underline{\mathcal{S}}\mathbf{x})(s, t) := \underline{S}_0(s)\mathbf{x}(s, t)$$

with  $\underline{S}_0(s) > 0$

*Proof.* See Appendix A.5. ■

**Lemma 2** (Primal Slope Restricted IQC). *Suppose that for  $z_\Delta \in \mathbf{L}_e, [a, b]$ ,  $(\Delta z_\Delta)(s, t) = \phi(z_\Delta(s, t))$  for all  $s \in [a, b]$  and  $t > 0$ , where  $\phi(0) = 0$  and  $\phi$  satisfies*

$$\hat{\alpha} \leq \frac{\phi(z_1) - \phi(z_2)}{z_1 - z_2} < \hat{\beta} \quad \text{for all } z_1, z_2 \in \mathbb{R}, \quad z_1 \neq z_2$$

with  $\hat{\beta} > 0$ . Then  $\Delta$  satisfies the primal Hard IQC defined by  $\Psi_\Delta = I$  and

$$\mathcal{V}_\Delta = \begin{bmatrix} \hat{\beta} & -1 \\ -\hat{\alpha} & 1 \end{bmatrix}^* \begin{bmatrix} 0 & \mathcal{S} \\ \mathcal{S} & 0 \end{bmatrix} \begin{bmatrix} \hat{\beta} & -1 \\ -\hat{\alpha} & 1 \end{bmatrix} \quad (10)$$

for any  $\mathcal{S} = \mathcal{P}_{\{S_0, S_1, S_2\}} \in \Pi_3$  with

$$S_0(s) := m(s) + \int_a^b H(s, \eta) d\eta,$$

$$S_1(s, \theta) = S_2(s, \theta) := -H(s, \theta),$$

with  $m(s) \geq 0$  and  $H(s, \theta) = H(\theta, s) \geq 0$  for all  $s, \theta \in [a, b]$ .

*Proof.* See Appendix A.6. ■

**Corollary 3** (Dual Slope Restricted IQC). *Suppose that for  $z_\Delta \in \mathbf{L}_e, [a, b]$ ,  $(\Delta z_\Delta)(s, t) = \phi(z_\Delta(s, t))$  for all  $s \in [a, b]$  and  $t > 0$ , where  $\phi(0) = 0$  and  $\phi$  satisfies*

$$\hat{\alpha} \leq \frac{\phi(z_1) - \phi(z_2)}{z_1 - z_2} < \hat{\beta} \quad \text{for all } z_1, z_2 \in \mathbb{R}, \quad z_1 \neq z_2$$

with  $\hat{\beta} > 0$ . Then  $\Delta$  satisfies the dual Hard IQC defined by  $\underline{\Psi}_\Delta = I$  and

$$\underline{\mathcal{V}}_\Delta = \begin{bmatrix} \hat{\beta} & -1 \\ -\hat{\alpha} & 1 \end{bmatrix}^* \begin{bmatrix} 0 & \underline{\mathcal{S}} \\ \underline{\mathcal{S}} & 0 \end{bmatrix} \begin{bmatrix} \hat{\beta} & -1 \\ -\hat{\alpha} & 1 \end{bmatrix} \quad (11)$$

for any  $\underline{\mathcal{S}} = \mathcal{P}_{\{\underline{S}_0, \underline{S}_1, \underline{S}_2\}} \in \Pi_3$  with

$$\underline{S}_0(s) := R_0(s) + \lambda, \quad \underline{S}_1(s, \theta) := \lambda, \quad \underline{S}_2(s, \theta) := \lambda$$

with  $R_0(s) > 0$  and  $\lambda \geq 0$ .

*Proof.* See Appendix A.7. ■

**Example 2 (Sine–Gordon Wave Equation).** Consider boundary control of the sine–Gordon equation shifted about the unstable equilibrium  $v = \pi$ :

$$\begin{aligned}\partial_t^2 \mathbf{v}(t, s) &= \partial_s^2 \mathbf{v}(t, s) - 3 \sin(\mathbf{v}(t, s)), \\ \mathbf{v}(t, 0) &= \pi, \quad \partial_s \mathbf{v}(t, 1) = x(t), \quad \dot{x}(t) = u(t).\end{aligned}$$

Let  $\mathbf{z}(t, s) = \mathbf{v}(t, s) - \pi$ . Then  $\sin(\mathbf{v}) = -\sin(\mathbf{z})$  and the shifted dynamics are

$$\begin{aligned}\partial_t^2 \mathbf{z}(t, s) &= \partial_s^2 \mathbf{z}(t, s) + 3\mathbf{z}(t, s) - 3w_\Delta(t, s), \\ \mathbf{z}(t, 0) &= 0, \quad \partial_s \mathbf{z}(t, 1) = x(t), \quad \dot{x}(t) = u(t), \\ z_\Delta(t, s) &= \mathbf{z}(t, s),\end{aligned}$$

The PIE realization of  $P$  has fundamental state  $\mathbf{x}_P(t) = [x(t) \ \partial_s \mathbf{z}(t) \ \partial_s^2 \mathbf{z}(t)]^\top$ , uncertainty  $(\Delta z_\Delta)(t, s) = \mathbf{z}(t, s) - \sin(\mathbf{z}(t, s))$ , and PI operators

$$\begin{aligned}\mathcal{T}_P &:= \Pi \left[ \begin{array}{c|c} 1 & [0 \ 0] \\ \hline \begin{bmatrix} 0 \\ s \end{bmatrix} & \left\{ \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -\theta \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & -s \end{bmatrix} \end{array} \right\}, \\ \mathcal{A}_P &:= \Pi \left[ \begin{array}{c|c} 0 & [0 \ 0] \\ \hline \begin{bmatrix} 3s \\ 0 \end{bmatrix} & \left\{ \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{5} & -3\theta \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -3s \\ 0 & 0 \end{bmatrix} \end{array} \right\}, \\ \mathcal{B}_P &:= \Pi \left[ \begin{array}{c|c} \emptyset & 0 \\ \hline \emptyset & \left\{ \begin{bmatrix} -3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{array} \right\}, \\ \mathcal{B}_u &:= \Pi \left[ \begin{array}{c|c} 1 & \emptyset \\ \hline \begin{bmatrix} 0 \\ 0 \end{bmatrix} & \{ \emptyset, \emptyset, \emptyset \} \end{array} \right], \\ \mathcal{C}_P &:= \Pi \left[ \begin{array}{c|c} \emptyset & \emptyset \\ \hline s & \{ [0 \ 0], [0 \ -\theta], [0 \ -s] \} \end{array} \right]\end{aligned}$$

Since no performance channels are included, we consider only robust stability. Accordingly, we maximize the stability margin  $\epsilon$  in (7), any solution with  $\epsilon > 0$  certifies robust stability. We perform two tests. The first uses the dual sector multiplier  $\underline{\mathcal{V}}_\Delta^{\text{sec}}$ , parameterized as in (9), and the second uses the dual slope multiplier  $\underline{\mathcal{V}}_\Delta^{\text{slope}}$ , parameterized as in (11). The resulting margins are  $\epsilon^{\text{sec}} = 1.1 \times 10^{-4}$  and  $\epsilon^{\text{slope}} = 6.5 \times 10^{-4}$ , respectively. Where the latter synthesized the following controller,

$$\begin{aligned}u(t) &= -16.55x(t) + \int_0^1 K_1(s) \partial_s \dot{\mathbf{z}}(t, s) ds \\ &\quad + \int_0^1 K_2(s) \partial_s^2 \mathbf{z}(t, s) ds, \\ K_1(s) &= 13.96s^{10} - 32.93s^9 + 21.90s^8 - 27.01s^7 \\ &\quad + 57.58s^6 - 12.80s^5 - 62.88s^4 + 71.53s^3 \\ &\quad - 42.35s^2 + 10.88s - 8.02, \\ K_2(s) &= 26.53s^{10} - 31.31s^9 - 282.70s^8 + 839.04s^7 \\ &\quad - 934.40s^6 + 474.54s^5 - 39.19s^4 - 58.83s^3 \\ &\quad - 21.15s^2 + 32.48s - 0.39.\end{aligned}$$

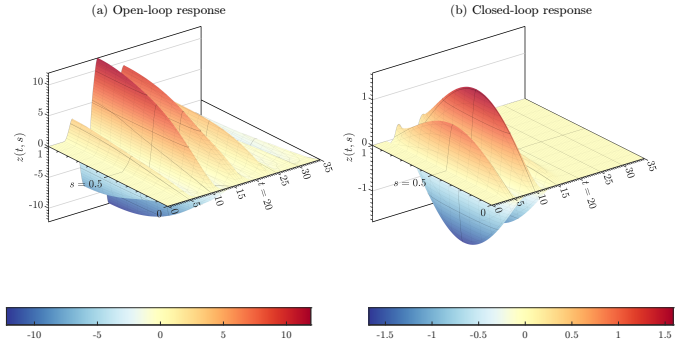
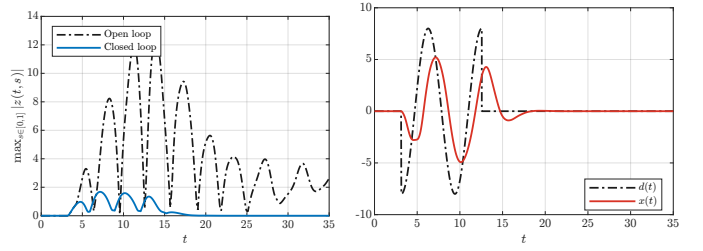


Fig. 6. Nonlinear simulation sine–Gordon: (a) open-loop response and (b) response under the slope-multiplier controller.



(a) Spatial-amplitude comparison (b) Disturbance and control effort

Fig. 7. Time-domain data corresponding to Fig. 6: (a) the spatial peak  $\max_{s \in [0,1]} |\mathbf{z}(t, s)|$  and (b) the injected disturbance  $d(t)$  and boundary control effort  $x(t)$ .

## VIII. CONCLUSION

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## APPENDIX A MATHEMATICAL PROOFS

### A. Proof of Theorem 1

Suppose that (2) holds. This implies,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} P_\Delta \\ I \end{bmatrix} w_\Delta, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} P_\Delta \\ I \end{bmatrix} w_\Delta \right\rangle_{\mathbf{L}_{[a,b]}} + \left\langle p_T \Psi_p \begin{bmatrix} P_{p\Delta} \\ 0 \end{bmatrix} w_\Delta, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} P_{p\Delta} \\ 0 \end{bmatrix} w_\Delta \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2.$$

By Definition 11 we can then infer,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} P_\Delta w_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} P_\Delta w_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2.$$

Then, using Theorem 6 in [11], we find that **robust stability** is guaranteed. Moreover, rewriting (2) as,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} + \left\langle p_T \Psi_p \begin{bmatrix} z_p \\ w_p \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} z_p \\ w_p \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon (\|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2 + \|p_T w_p\|_{\mathbf{L}_{[a,b]}}^2),$$

and given that  $\Psi_\Delta$  and  $\mathcal{M}_{\mathcal{V}_\Delta}$  satisfy the IQC in Definition 10 leads to,

$$\left\langle p_T \Psi_p \begin{bmatrix} z_p \\ w_p \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} z_p \\ w_p \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_p\|_{\mathbf{L}_{[a,b]}}^2.$$

Thus by Definition 11 **robust performance** is achieved.

### B. Proof of Theorem 3

1)  $\implies$  2) Assume there exists a  $0 < \rho < 1$  such that  $\|p_T G_0 \tilde{w}\|_{\mathbf{L}} \leq \rho \|p_T \tilde{w}\|_{\mathbf{L}}$  for all  $\tilde{w} \in \mathbf{L}$  and  $T \geq 0$ . We need to show that  $\mathcal{I} \leq -\epsilon \|p_T w\|_{\mathbf{L}}^2$  for some  $\epsilon > 0$ . First since  $\Gamma$  is causally boundedly invertible,  $\Theta \begin{bmatrix} P \\ I \end{bmatrix} w = \begin{bmatrix} G_0 \\ I \end{bmatrix} \Gamma w$ , and hence  $\mathcal{I} = \|p_T G_0 \Gamma w\|_{\mathbf{L}}^2 - \|p_T \Gamma w\|_{\mathbf{L}}^2$ . By statement 1 and causality of  $G_0$ ,

$$\mathcal{I} \leq -(1 - \rho^2) \|p_T \Gamma w\|_{\mathbf{L}}^2 \leq -\frac{1 - \rho^2}{\|\Gamma^{-1}\|_{\mathcal{L}(\mathbf{L})}^2} \|p_T w\|_{\mathbf{L}}^2,$$

where the second inequality follows from  $\|p_T w\|_{\mathbf{L}} = \|p_T \Gamma^{-1} p_T \Gamma w\|_{\mathbf{L}} \leq \|\Gamma^{-1}\|_{\mathcal{L}(\mathbf{L})} \|p_T \Gamma w\|_{\mathbf{L}}$ . Thus,  $\epsilon = \frac{(1 - \rho^2)}{\|\Gamma^{-1}\|_{\mathcal{L}(\mathbf{L})}^2} > 0$ .

2)  $\implies$  1) Assume that there exists a  $\epsilon > 0$  such that  $\mathcal{I} \leq -\epsilon \|p_T w\|_{\mathbf{L}}^2$  for every  $w$ , and every  $T \geq 0$ . We need to establish there exists a  $0 < \rho < 1$  such that  $\|p_T G_0 \tilde{w}\|_{\mathbf{L}} \leq \rho \|p_T \tilde{w}\|_{\mathbf{L}}$  for all  $\tilde{w} \in \mathbf{L}$  and  $T \geq 0$ . For arbitrary  $\tilde{w}$ , set  $w = \Gamma^{-1} \tilde{w}$ . Then

$$\|p_T G_0 \tilde{w}\|_{\mathbf{L}}^2 - \|p_T \tilde{w}\|_{\mathbf{L}}^2 \leq -\epsilon \|p_T \Gamma^{-1} \tilde{w}\|_{\mathbf{L}}^2.$$

Since  $\Gamma$  is bounded and causal,  $\|p_T \tilde{w}\|_{\mathbf{L}} \leq \|\Gamma\|_{\mathcal{L}(\mathbf{L})} \|p_T \Gamma^{-1} \tilde{w}\|_{\mathbf{L}}$ . Therefore,

$$\|p_T G_0 \tilde{w}\|_{\mathbf{L}}^2 \leq \left(1 - \frac{\epsilon}{\|\Gamma\|_{\mathcal{L}(\mathbf{L})}^2}\right) \|p_T \tilde{w}\|_{\mathbf{L}}^2.$$

Choose  $0 < \rho < 1$  such that  $1 - \epsilon/\|\Gamma\|^2 \leq \rho^2 < 1$ , then  $\|p_T G_0 \tilde{w}\|_{\mathbf{L}} \leq \rho \|p_T \tilde{w}\|_{\mathbf{L}}$  for every  $\tilde{w}$  and  $T \geq 0$ .

1)  $\iff$  3) We show that there exists a causally boundedly invertible  $\underline{\Gamma}$  such that

$$\mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} G_0^\top \\ I \end{bmatrix} \underline{\Gamma} w.$$

then we can apply the previous arguments to show 1)  $\iff$  3). Define

$$S := \Theta_{11} - \Theta_{12} \Theta_{22}^{-1} \Theta_{21}.$$

Since  $\Theta$  and  $\Theta_{22}$  are boundedly invertible, so is  $S$  by Schur complement. Block inversion of  $\mathbf{D}(\Theta)$  gives

$$\mathbf{D}(\Theta)_{21} = (S^\top)^{-1} \Theta_{21}^\top (\Theta_{22}^\top)^{-1}, \quad \mathbf{D}(\Theta)_{22} = (S^\top)^{-1}.$$

Therefore, define

$$\underline{\Gamma} := \mathbf{D}(\Theta)_{21} P^\top + \mathbf{D}(\Theta)_{22} = (S^\top)^{-1} (I + P \Theta_{22}^{-1} \Theta_{21})^\top.$$

Now  $\Gamma = \Theta_{22} (I + \Theta_{22}^{-1} \Theta_{21} P)$ . Thus, invertibility of  $\Gamma$  and the push-through rule imply that  $I + P \Theta_{22}^{-1} \Theta_{21}$ , and hence  $\underline{\Gamma}$ , is causally boundedly invertible. Normalizing the upper block gives

$$\mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} G_0^\top \\ I \end{bmatrix} \underline{\Gamma} w, \quad \underline{G}_0 := (\Theta_{22}^\top)^{-1} (P^\top \underline{\Gamma}^{-1} + \Theta_{12}^\top).$$

Using

$$\underline{\Gamma}^{-\top} = S(I + P\Theta_{22}^{-1}\Theta_{21})^{-1}, \quad (I + P\Theta_{22}^{-1}\Theta_{21})^{-1}P\Theta_{22}^{-1} = P\Gamma^{-1},$$

where the second identity is the push-through rule, yields

$$\begin{aligned} \underline{G}_0 &= (\Theta_{22}^\top)^{-1}(P^\top \underline{\Gamma}^{-1} + \Theta_{12}^\top) \\ &= \left( \underline{\Gamma}^{-\top} P \Theta_{22}^{-1} + \Theta_{12} \Theta_{22}^{-1} \right)^\top \\ &= \left[ S(I + P\Theta_{22}^{-1}\Theta_{21})^{-1} P \Theta_{22}^{-1} + \Theta_{12} \Theta_{22}^{-1} \right]^\top \\ &= \left[ S P \Gamma^{-1} + \Theta_{12} \Theta_{22}^{-1} \right]^\top \\ &= \left[ (S P + \Theta_{12} \Theta_{22}^{-1} \Gamma) \Gamma^{-1} \right]^\top. \end{aligned}$$

We now show that the first part reduces to

$$\begin{aligned} S P + \Theta_{12} \Theta_{22}^{-1} \Gamma &= \Theta_{11} P - \Theta_{12} \Theta_{22}^{-1} \Theta_{21} P \\ &\quad + \Theta_{12} \Theta_{22}^{-1} \Theta_{21} P + \Theta_{12} \\ &= \Theta_{11} P + \Theta_{12}. \end{aligned}$$

Consequently,

$$\underline{G}_0 = [(\Theta_{11} P + \Theta_{12}) \Gamma^{-1}]^\top = G_0^\top,$$

and hence

$$\mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} G_0^\top \\ I \end{bmatrix} \Gamma w.$$

By Theorem 2,  $G_0$  and  $G_0^\top$  have the same induced  $L_2$ -gain. Hence, for any  $\rho > 0$ ,

$$\|p_T G_0 \tilde{w}\|_{\mathbf{L}} \leq \rho \|p_T \tilde{w}\|_{\mathbf{L}} \quad \text{for all } \tilde{w} \text{ and } T \geq 0 \quad \Longleftrightarrow \quad \|p_T G_0^\top \tilde{w}\|_{\mathbf{L}} \leq \rho \|p_T \tilde{w}\|_{\mathbf{L}} \quad \text{for all } \tilde{w} \text{ and } T \geq 0.$$

Since  $\underline{\Gamma}$  is causally boundedly invertible, the same arguments used for 1)  $\implies$  2) and 2)  $\implies$  1), with  $G_0$ ,  $\Gamma$ ,  $w$ , and  $\underline{\mathcal{I}}$  replaced by  $G_0^\top$ ,  $\underline{\Gamma}$ ,  $\underline{w}$ , and  $\underline{\mathcal{I}}$ , respectively, establish 1)  $\iff$  3).

a) *Realization of  $G_0$ .*: Let  $\mathbf{x}$  collect the plant, filter, and any controller states, so that the controller is already included in the primal operators. Partition the filtered realization as

$$\begin{bmatrix} \mathcal{T} \dot{\mathbf{x}} \\ \tilde{z} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C}_z & \mathcal{D}_z \\ \mathcal{C}_w & \mathcal{D}_w \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ w \end{bmatrix}.$$

If  $\mathcal{D}_w$  is invertible, eliminating  $w$  gives the realization

$$G_0 \sim \{\mathcal{T}, \mathcal{A} - \mathcal{B} \mathcal{D}_w^{-1} \mathcal{C}_w, \mathcal{B} \mathcal{D}_w^{-1}, \mathcal{C}_z - \mathcal{D}_z \mathcal{D}_w^{-1} \mathcal{C}_w, \mathcal{D}_z \mathcal{D}_w^{-1}\}, \quad (12)$$

with input  $\tilde{w}$  and output  $\tilde{z}$ .

### C. Proof of Theorem 1

If the primal LPI holds, the standard KYP integration gives the hard IQC

$$\left\langle p_T \Psi \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_\Psi \Psi \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2.$$

Theorem 1 then gives the result. It remains to consider the dual LPI. Define  $V(\mathbf{x}) = \langle \mathcal{T} \mathbf{x}, \mathcal{P} \mathcal{T} \mathbf{x} \rangle_{\mathbf{L}_{[a,b]}} \geq 0$ . For any  $\underline{w} \in \mathbf{L}_{\mathbf{e},[a,b]}^{\mathbf{n}_z}$ , suppose  $\underline{v}$  satisfies (5) for some  $\underline{\mathbf{x}} \in \mathbf{L}_{\mathbf{e},[a,b]}^{\mathbf{n}_x}$ . By (5), we have

$$\left\langle \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix}, \begin{bmatrix} \mathcal{T}^* \mathcal{P} \mathcal{A} + (*) & \mathcal{T}^* \mathcal{P} \mathcal{B} \\ \mathcal{B}^* \mathcal{P} \mathcal{T} & \epsilon I \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix} \right\rangle + \langle \underline{v}(t), \underline{\mathcal{V}} \underline{v}(t) \rangle \leq 0.$$

After integrating in time, we obtain

$$\epsilon \|p_T \underline{w}\|_{\mathbf{L}_{[a,b]}}^2 + \langle p_T \underline{v}, p_T \underline{\mathcal{V}} \underline{v} \rangle_{\mathbf{L}_{[a,b]}} \leq -V(\underline{\mathbf{x}}(T)) + V(\underline{\mathbf{x}}(0)).$$

Since  $\underline{\mathbf{x}}(0) = 0$  and  $V(\underline{\mathbf{x}}(T)) \geq 0$ ,

$$\left\langle p_T \underline{\Psi} \begin{bmatrix} \underline{z} \\ \underline{w} \end{bmatrix}, p_T \mathcal{M}_{\underline{\Psi}} \underline{\Psi} \begin{bmatrix} \underline{z} \\ \underline{w} \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T \underline{w}\|_{\mathbf{L}_{[a,b]}}^2.$$

Using the  $(\mathbf{D}(\Theta), J_{\mathbf{n}_w, \mathbf{n}_z})$  factorization, this is

$$\left\langle p_T \mathbf{D}(\Theta) \begin{bmatrix} \underline{z} \\ \underline{w} \end{bmatrix}, p_T J_{\mathbf{n}_w, \mathbf{n}_z} \mathbf{D}(\Theta) \begin{bmatrix} \underline{z} \\ \underline{w} \end{bmatrix} \right\rangle_{\mathbf{L}_{[a, b]}} \leq -\epsilon \|p_T \underline{w}\|_{\mathbf{L}_{[a, b]}}^2.$$

By Theorem ??, there exists  $\epsilon > 0$  such that

$$\left\langle p_T \Theta \begin{bmatrix} z \\ w \end{bmatrix}, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \Theta \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a, b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a, b]}}^2.$$

Using the  $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$  factorization gives the primal hard IQC, and Theorem 1 gives robust stability and robust performance.

#### D. Proof of Theorem 2

Since  $\mathcal{P} \succ 0$ , define  $\mathcal{K}_P$  and  $\mathcal{K}_\Psi$  by

$$\mathcal{Z}^* = [\mathcal{K}_P \quad \mathcal{K}_\Psi] \mathcal{P}.$$

Set

$$\underline{\mathcal{A}} := \underline{\mathcal{A}}_0 + \begin{bmatrix} \mathcal{K}_P^* \\ \mathcal{K}_\Psi^* \end{bmatrix} [\mathcal{B}_u^* \quad 0], \quad \underline{\mathcal{B}} := \underline{\mathcal{B}}_0 + \begin{bmatrix} \mathcal{K}_P^* \\ \mathcal{K}_\Psi^* \end{bmatrix} \mathcal{D}_{zu}^*.$$

Then (7) is the dual KYP inequality for

$$\mathbf{D}(\Psi) \begin{bmatrix} P_K^\top \\ I \end{bmatrix} \underline{w} := \left\{ \begin{bmatrix} \mathcal{T}\dot{\mathbf{x}}(t) \\ \underline{v}(t) \end{bmatrix} = \begin{bmatrix} \underline{\mathcal{A}} & \underline{\mathcal{B}} \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \underline{w}(t) \end{bmatrix}, \quad P_K := K \star P, \quad \underline{\mathbf{x}} := \begin{bmatrix} \underline{\mathbf{x}}_P \\ \underline{\mathbf{x}}_\Psi \end{bmatrix} \right\}.$$

For

$$V(\underline{\mathbf{x}}) := \langle \underline{\mathcal{T}}\underline{\mathbf{x}}, \mathcal{P}\underline{\mathcal{T}}\underline{\mathbf{x}} \rangle \geq 0,$$

the standard storage argument gives, for zero initial conditions,

$$\left\langle p_T \mathbf{D}(\Psi) \begin{bmatrix} P_K^\top \\ I \end{bmatrix} \underline{w}, p_T \mathcal{M}_\Psi \mathbf{D}(\Psi) \begin{bmatrix} P_K^\top \\ I \end{bmatrix} \underline{w} \right\rangle_{\mathbf{L}_{[a, b]}} \leq -\epsilon \|p_T \underline{w}\|_{\mathbf{L}_{[a, b]}}^2.$$

Define

$$\Theta := \mathcal{M}_\mathcal{U} \Psi.$$

Using

$$\underline{\mathcal{V}} = \mathbf{D}(\mathcal{U})^* J_{\mathbf{n}_w, \mathbf{n}_z} \mathbf{D}(\mathcal{U}), \quad \mathbf{D}(\Theta) = \mathcal{M}_{\mathbf{D}(\mathcal{U})} \mathbf{D}(\Psi),$$

the preceding inequality becomes

$$\left\langle p_T \mathbf{D}(\Theta) \begin{bmatrix} P_K^\top \\ I \end{bmatrix} \underline{w}, p_T J_{\mathbf{n}_w, \mathbf{n}_z} \mathbf{D}(\Theta) \begin{bmatrix} P_K^\top \\ I \end{bmatrix} \underline{w} \right\rangle_{\mathbf{L}_{[a, b]}} \leq -\epsilon \|p_T \underline{w}\|_{\mathbf{L}_{[a, b]}}^2. \quad (13)$$

By Lemma ??, there exist invertible coordinate operators  $\mathcal{M}_{\Gamma_K}$  and  $\mathcal{M}_{\Gamma_K}$  such that

$$\begin{aligned} \Theta \begin{bmatrix} P_K \\ I \end{bmatrix} w &= \begin{bmatrix} P_K \star (\Theta^\dagger R) \\ I \end{bmatrix} \mathcal{M}_{\Gamma_K} w, \\ \mathbf{D}(\Theta) \begin{bmatrix} P_K^\top \\ I \end{bmatrix} \underline{w} &= \begin{bmatrix} P_K^\top \star (R(\Theta^\dagger)^\top) \\ I \end{bmatrix} \mathcal{M}_{\Gamma_K} \underline{w}, \end{aligned}$$

where

$$\mathbf{D}(\Theta)^\dagger R = R(\Theta^\dagger)^\top, \quad P_K^\top \star (R(\Theta^\dagger)^\top) = (P_K \star (\Theta^\dagger R))^\top.$$

Thus (13) is the dual feasibility condition associated with the primal filtered system. Theorem ?? therefore gives some  $\epsilon > 0$  such that

$$\left\langle p_T \Theta \begin{bmatrix} P_K \\ I \end{bmatrix} w, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \Theta \begin{bmatrix} P_K \\ I \end{bmatrix} w \right\rangle_{\mathbf{L}_{[a, b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a, b]}}^2.$$

Since  $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$  is the primal hard-IQC factorization, Theorem 1 gives robust stability and robust performance of  $K \star P \star \Delta$ .

### E. Proof of Lemma 1

Suppose that the dual IQC

$$\left\langle p_T \begin{bmatrix} P_\Delta^\top \underline{w}_\Delta \\ \underline{w}_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \begin{bmatrix} P^\top \underline{w}_\Delta \\ \underline{w}_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}}$$

is parametrized according to (9). We need to show that  $\Delta$  satisfies the primal IQC. First, observe the dual IQC can be factorized as

$$\mathcal{V}_\Delta = \frac{1}{2} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}^* \begin{bmatrix} \mathcal{S} & I \\ \mathcal{S} & -I \end{bmatrix}^* \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} \underbrace{\begin{bmatrix} \mathcal{S} & I \\ \mathcal{S} & -I \end{bmatrix}}_{\Theta_1} \underbrace{\begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}}_{\Theta_2} \quad (14)$$

Then, we use the multiplicative and involutive property of the dual operation such that  $\mathbf{D}(\Theta_1 \Theta_2) = \mathbf{D}(\Theta_1) \mathbf{D}(\Theta_2)$  for any two invertible  $\Theta_1$  and  $\Theta_2$  and  $\mathbf{D}(\mathbf{D}(\Theta)) = \Theta$  to recover the primal IQC. Since  $\underline{S}_0 \succeq 0$ ,  $\underline{\mathcal{S}}$  is closed under inversion with  $\underline{S}_0^{-1} \succeq 0$ . Applying these properties to the two factors of  $\underline{\Theta}$  in (14) gives

$$\begin{aligned} \Theta &= \mathbf{D}(\underline{\Theta}) = \mathbf{D}\left(\frac{1}{\sqrt{2}} \begin{bmatrix} I & \underline{\mathcal{S}} \\ I & -\underline{\mathcal{S}} \end{bmatrix}\right) \mathbf{D}\left(\begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}\right) \\ &= \frac{1}{\sqrt{2}(\beta - \alpha)} \begin{bmatrix} -\underline{\mathcal{S}}^{-1} & -I \\ -\underline{\mathcal{S}}^{-1} & I \end{bmatrix} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}. \end{aligned}$$

We omit  $\frac{1}{(\beta - \alpha)} > 0$  therefore,

$$\mathcal{V}_\Delta = \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}^* \begin{bmatrix} 0 & \underline{\mathcal{S}}^{-1} \\ \underline{\mathcal{S}}^{-1} & 0 \end{bmatrix} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}$$

For completion we use the proof in [11] Then, for all  $s \in [a, b]$  and  $t > 0$ , we have that

$$(\beta z_\Delta(s, t) - \phi(z_\Delta(s, t)))(\phi(z_\Delta(s, t)) - \alpha z_\Delta(s, t)) \geq 0,$$

and hence

$$(\beta z_\Delta(s, t) - \phi(z_\Delta(s, t)))\underline{S}_0^{-1}(s)(\phi(z_\Delta(s, t)) - \alpha z_\Delta(s, t)) \geq 0.$$

Therefore,  $\Delta$  satisfies the IQC defined by  $\underline{\Psi}_\Delta = I$  and (9).

### F. Proof of the Primal Slope Restricted IQC

Suppose that the primal IQC

$$\left\langle p_T \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}}$$

is parametrized according to (10). We need to show that  $\Delta$  satisfies the primal IQC. Define

$$p(s, t) := \hat{\beta} z_\Delta(s, t) - \phi(z_\Delta(s, t)), \quad q(s, t) := \phi(z_\Delta(s, t)) - \hat{\alpha} z_\Delta(s, t).$$

Applying the slope restriction to  $z_\Delta(s, t)$  and 0, and using  $\phi(0) = 0$ , gives

$$p(s, t)q(s, t) \geq 0.$$

Likewise, applying it to  $z_\Delta(s, t)$  and  $z_\Delta(\theta, t)$  gives

$$(p(s, t) - p(\theta, t))(q(s, t) - q(\theta, t)) \geq 0.$$

Using the 3-PI representation of  $\mathcal{S}$ , we obtain

$$\begin{aligned} (\mathcal{S}x)(s) &= S_0(s)x(s) + \int_a^s S_1(s, \theta)x(\theta)d\theta + \int_s^b S_2(s, \theta)x(\theta)d\theta \\ &= \left(m(s) + \int_a^b H(s, \eta)d\eta\right)x(s) - \int_a^s H(s, \theta)x(\theta)d\theta - \int_s^b H(s, \theta)x(\theta)d\theta \\ &= m(s)x(s) + \int_a^b H(s, \theta)(x(s) - x(\theta))d\theta. \end{aligned}$$

Hence, using  $H(s, \theta) = H(\theta, s)$ ,

$$\left\langle \begin{bmatrix} p \\ q \end{bmatrix}, \begin{bmatrix} 0 & \mathcal{S} \\ \mathcal{S} & 0 \end{bmatrix} \begin{bmatrix} p \\ q \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} = 2 \int_a^b m(s)p(s, t)q(s, t)ds + \int_a^b \int_a^b H(s, \theta)(p(s, t) - p(\theta, t))(q(s, t) - q(\theta, t))d\theta ds \geq 0.$$

Integrating this inequality over  $t \in [0, T]$  proves the IQC for every  $T \geq 0$ , and hence proves the stated primal hard IQC.

### G. Proof of the Dual Slope Restricted IQC

Suppose that the dual IQC

$$\left\langle p_T \begin{bmatrix} P_\Delta^\top \underline{w}_\Delta \\ \underline{w}_\Delta \end{bmatrix}, p_T \mathcal{M}_{\underline{\Psi}_\Delta} \begin{bmatrix} P_\Delta^\top \underline{w}_\Delta \\ \underline{w}_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}}$$

is parametrized according to (11). We need to show that  $\Delta$  satisfies the primal IQC. First, the 3-PI coefficients in the lemma give

$$(\underline{\mathcal{S}}x)(s) = (R_0(s) + \lambda)x(s) + \lambda \int_a^b x(\theta)d\theta.$$

Let

$$r(s) := (R_0(s) + \lambda)^{-1}, \quad R := \int_a^b r(\eta)d\eta.$$

To compute the inverse, set  $y = \underline{\mathcal{S}}x$  and  $\mu := \int_a^b x(\theta)d\theta$ . Then, by the preceding expression for  $\underline{\mathcal{S}}$ ,

$$y(s) = (R_0(s) + \lambda)x(s) + \lambda\mu.$$

Thus,  $(R_0(s) + \lambda)x(s) = y(s) - \lambda\mu$ . Since  $R_0(s) > 0$  and  $\lambda \geq 0$ ,  $R_0(s) + \lambda$  is invertible, and multiplication by  $r(s) = (R_0(s) + \lambda)^{-1}$  gives

$$x(s) = r(s)(y(s) - \lambda\mu).$$

Integrating both sides over  $s$  gives

$$\begin{aligned} \mu &= \int_a^b r(s)y(s)ds - \lambda\mu \int_a^b r(s)ds, \\ (1 + \lambda R)\mu &= \int_a^b r(s)y(s)ds, \end{aligned}$$

and therefore

$$\mu = \frac{\int_a^b r(\theta)y(\theta)d\theta}{1 + \lambda R}.$$

Substitution into the expression for  $x$  yields

$$(\underline{\mathcal{S}}^{-1}y)(s) = r(s)y(s) - \frac{\lambda r(s)}{1 + \lambda R} \int_a^b r(\theta)y(\theta)d\theta. \quad (15)$$

Now define

$$H(s, \theta) := \frac{\lambda}{1 + \lambda R} r(s)r(\theta) \geq 0, \quad m(s) := \frac{r(s)}{1 + \lambda R} > 0.$$

Both are nonnegative, and

$$m(s) + \int_a^b H(s, \theta)d\theta = \frac{r(s)}{1 + \lambda R} + \frac{\lambda R r(s)}{1 + \lambda R} = r(s).$$

Consequently,

$$\begin{aligned} m(s)y(s) + \int_a^b H(s, \theta)(y(s) - y(\theta))d\theta \\ = r(s)y(s) - \frac{\lambda r(s)}{1 + \lambda R} \int_a^b r(\theta)y(\theta)d\theta, \end{aligned}$$

which is exactly (15). Thus  $\underline{\mathcal{S}}^{-1}$  has the primal slope form. Then we can apply the arguments from the preceding proofs to conclude that  $\Delta$  satisfies the IQC defined by  $\underline{\Psi}_\Delta = I$  and (11).